SQL Coursework
Financial Services Database

Part 1: Database Setup And Sample Data

– Step 1: Create the Database

```
CREATE DATABASE FinanceHub;
USE FinanceHub;
```

– Step 2: Create Tables

```
-- Customers table - Bank customers information
CREATE TABLE Customers (
    customer_id INT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    phone VARCHAR(20),
    date_of_birth DATE,
    address VARCHAR(200),
    city VARCHAR(50),
    state VARCHAR(50),
    country VARCHAR(50),
    zip_code VARCHAR(20),
    registration_date DATE NOT NULL,
    customer_type VARCHAR(30) DEFAULT 'Individual',
    risk_profile VARCHAR(20) DEFAULT 'Medium',
    credit_score INT,
    annual_income DECIMAL(12, 2),
    kyc_verified BOOLEAN DEFAULT FALSE
);
-- Accounts table - Bank accounts
CREATE TABLE Accounts (
    account_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    account_number VARCHAR(20) UNIQUE NOT NULL,
    account_type VARCHAR(30) NOT NULL,
    account_status VARCHAR(20) DEFAULT 'Active',
    opening_date DATE NOT NULL,
    current_balance DECIMAL(15, 2) DEFAULT 0.00,
    currency VARCHAR(10) DEFAULT 'USD',
    interest_rate DECIMAL(5, 2) DEFAULT 0.00,
    minimum_balance DECIMAL(12, 2) DEFAULT 0.00,
    branch_code VARCHAR(20),
    last_transaction_date DATE,
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);
```

```
-- Transactions table - All financial transactions
CREATE TABLE Transactions (
    transaction_id INT PRIMARY KEY,
    account_id INT NOT NULL,
    transaction_type VARCHAR(30) NOT NULL,
    transaction_date DATETIME NOT NULL,
    amount DECIMAL(15, 2) NOT NULL,
    balance_after DECIMAL(15, 2),
    description VARCHAR(200),
    reference_number VARCHAR(50) UNIQUE,
    transaction_status VARCHAR(20) DEFAULT 'Completed',
    channel VARCHAR(30),
    category VARCHAR(50),
    FOREIGN KEY (account_id) REFERENCES Accounts(account_id)
);
```

```
-- Loans table - Loan information
CREATE TABLE Loans (
    loan_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    loan_type VARCHAR(30) NOT NULL,
    loan_amount DECIMAL(15, 2) NOT NULL,
    interest_rate DECIMAL(5, 2) NOT NULL,
    tenure_months INT NOT NULL,
    monthly_emi DECIMAL(12, 2) NOT NULL,
    outstanding_balance DECIMAL(15, 2) NOT NULL,
    disbursement_date DATE NOT NULL,
    maturity_date DATE NOT NULL,
    loan_status VARCHAR(20) DEFAULT 'Active',
    collateral_type VARCHAR(50),
    collateral_value DECIMAL(15, 2),
    purpose VARCHAR(100),
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);

-- LoanPayments table - Loan EMI payments
CREATE TABLE LoanPayments (
    payment_id INT PRIMARY KEY,
    loan_id INT NOT NULL,
    payment_date DATE NOT NULL,
    amount_paid DECIMAL(12, 2) NOT NULL,
    principal_amount DECIMAL(12, 2) NOT NULL,
    interest_amount DECIMAL(12, 2) NOT NULL,
    outstanding_balance DECIMAL(15, 2) NOT NULL,
    payment_status VARCHAR(20) DEFAULT 'Paid',
    payment_method VARCHAR(30),
    late_fee DECIMAL(8, 2) DEFAULT 0.00,
    FOREIGN KEY (loan_id) REFERENCES Loans(loan_id)
);

-- CreditCards table - Credit card information
CREATE TABLE CreditCards (
    card_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    card_number VARCHAR(20) UNIQUE NOT NULL,
    card_type VARCHAR(30) NOT NULL,
    credit_limit DECIMAL(12, 2) NOT NULL,
    available_credit DECIMAL(12, 2) NOT NULL,
    current_balance DECIMAL(12, 2) DEFAULT 0.00,
    interest_rate DECIMAL(5, 2) NOT NULL,
    issue_date DATE NOT NULL,
    expiry_date DATE NOT NULL,
    card_status VARCHAR(20) DEFAULT 'Active',
    reward_points INT DEFAULT 0,
    annual_fee DECIMAL(8, 2) DEFAULT 0.00,
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);
```

```
-- CardTransactions table - Credit card transactions
CREATE TABLE CardTransactions (
    card_transaction_id INT PRIMARY KEY,
    card_id INT NOT NULL,
    transaction_date DATETIME NOT NULL,
    merchant_name VARCHAR(100),
    merchant_category VARCHAR(50),
    amount DECIMAL(12, 2) NOT NULL,
    transaction_type VARCHAR(20) NOT NULL,
    transaction_status VARCHAR(20) DEFAULT 'Posted',
    foreign_transaction BOOLEAN DEFAULT FALSE,
    reward_points_earned INT DEFAULT 0,
    FOREIGN KEY (card_id) REFERENCES CreditCards(card_id)
);

-- Investments table - Investment portfolios
CREATE TABLE Investments (
    investment_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    investment_type VARCHAR(50) NOT NULL,
    security_name VARCHAR(100),
    quantity DECIMAL(15, 4),
    purchase_price DECIMAL(15, 2) NOT NULL,
    current_price DECIMAL(15, 2),
    purchase_date DATE NOT NULL,
    maturity_date DATE,
    investment_status VARCHAR(20) DEFAULT 'Active',
    risk_level VARCHAR(20),
    returns_percentage DECIMAL(8, 2),
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);

-- Branches table - Bank branches
CREATE TABLE Branches (
    branch_id INT PRIMARY KEY,
    branch_code VARCHAR(20) UNIQUE NOT NULL,
    branch_name VARCHAR(100) NOT NULL,
    address VARCHAR(200),
    city VARCHAR(50),
    state VARCHAR(50),
    country VARCHAR(50),
    phone VARCHAR(20),
    manager_name VARCHAR(100),
    opening_date DATE,
    branch_type VARCHAR(30),
    total_customers INT DEFAULT 0,
    is_active BOOLEAN DEFAULT TRUE
);
```

```
-- Employees table - Bank employees
CREATE TABLE Employees (
    employee_id INT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    phone VARCHAR(20),
    hire_date DATE NOT NULL,
    job_title VARCHAR(100) NOT NULL,
    department VARCHAR(50) NOT NULL,
    branch_id INT,
    manager_id INT,
    salary DECIMAL(12, 2),
    performance_rating DECIMAL(3, 2),
    is_active BOOLEAN DEFAULT TRUE,
    FOREIGN KEY (branch_id) REFERENCES Branches(branch_id),
    FOREIGN KEY (manager_id) REFERENCES Employees(employee_id)
);

-- CustomerService table - Customer service interactions
CREATE TABLE CustomerService (
    ticket_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    employee_id INT,
    ticket_date DATETIME NOT NULL,
    issue_category VARCHAR(50) NOT NULL,
    issue_description TEXT,
    priority VARCHAR(20) DEFAULT 'Medium',
    status VARCHAR(20) DEFAULT 'Open',
    resolution_date DATETIME,
    satisfaction_rating INT CHECK (satisfaction_rating BETWEEN 1 AND 5),
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
    FOREIGN KEY (employee_id) REFERENCES Employees(employee_id)
);

-- FraudAlerts table - Fraud detection alerts
CREATE TABLE FraudAlerts (
    alert_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    account_id INT,
    alert_date DATETIME NOT NULL,
    alert_type VARCHAR(50) NOT NULL,
    risk_score INT CHECK (risk_score BETWEEN 0 AND 100),
    description TEXT,
    status VARCHAR(20) DEFAULT 'Pending',
    investigated_by INT,
    resolution_date DATETIME,
    is_fraud_confirmed BOOLEAN,
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id),
    FOREIGN KEY (account_id) REFERENCES Accounts(account_id),
    FOREIGN KEY (investigated_by) REFERENCES Employees(employee_id)
);
```

```
-- Tools table - Tools used by analysts
CREATE TABLE Tools (
    tool_id INT PRIMARY KEY,
    tool_name VARCHAR(100) NOT NULL,
    tool_category VARCHAR(50),
    cost DECIMAL(10, 2),
    license_type VARCHAR(50),
    version VARCHAR(20),
    is_active BOOLEAN DEFAULT TRUE
);

-- AnalystTools table - Junction table for analysts and their tools
CREATE TABLE AnalystTools (
    employee_id INT NOT NULL,
    tool_id INT NOT NULL,
    proficiency_level VARCHAR(20),
    is_primary_tool BOOLEAN DEFAULT FALSE,
    last_used_date DATE,
    PRIMARY KEY (employee_id, tool_id),
    FOREIGN KEY (employee_id) REFERENCES Employees(employee_id),
    FOREIGN KEY (tool_id) REFERENCES Tools(tool_id)
);

-- TrainingPrograms table - Training programs available
CREATE TABLE TrainingPrograms (
    program_id INT PRIMARY KEY,
    program_name VARCHAR(100) NOT NULL,
    program_category VARCHAR(50),
    duration_hours INT,
    cost DECIMAL(10, 2),
    provider VARCHAR(100),
    start_date DATE,
    end_date DATE
);

-- AnalystTraining table - Junction table for analysts and training
CREATE TABLE AnalystTraining (
    training_record_id INT PRIMARY KEY,
    employee_id INT NOT NULL,
    program_id INT NOT NULL,
    completion_date DATE,
    score INT CHECK (score BETWEEN 0 AND 100),
    is_certified BOOLEAN DEFAULT FALSE,
    FOREIGN KEY (employee_id) REFERENCES Employees(employee_id),
    FOREIGN KEY (program_id) REFERENCES TrainingPrograms(program_id)
);
```

-- Step 3: Insert Sample Data

-- Insert Customers

INSERT INTO Customers VALUES

(1001, 'John', 'Anderson', 'john.anderson@email.com', '555-0101', '1985-03-15', '123 Park Ave', 'New York', 'NY', 'USA', '10001', '2020-01-15', 'Individual', 'Low', 720, 95000.00, TRUE),

(1002, 'Sarah', 'Williams', 'sarah.w@email.com', '555-0102', '1990-07-22', '456 Market St', 'San Francisco', 'CA', 'USA', '94102', '2020-02-20', 'Individual', 'Medium', 680, 78000.00, TRUE),

(1003, 'Michael', 'Brown', 'michael.b@email.com', '555-0103', '1978-11-10', '789 Main St', 'Chicago', 'IL', 'USA', '60601', '2020-03-18', 'Individual', 'Medium', 695, 120000.00, TRUE),

(1004, 'Emily', 'Davis', 'emily.davis@email.com', '555-0104', '1988-05-30', '321 Oak Rd', 'Boston', 'MA', 'USA', '02101', '2020-04-12', 'Individual', 'High', 750, 150000.00, TRUE),

(1005, 'Robert', 'Johnson', 'robert.j@email.com', '555-0105', '1982-09-14', '654 Pine St', 'Seattle', 'WA', 'USA', '98101', '2020-05-25', 'Individual', 'Low', 710, 88000.00, TRUE),

(1006, 'Jennifer', 'Martinez', 'jennifer.m@email.com', '555-0106', '1992-12-08', '987 Elm Ave', 'Miami', 'FL', 'USA', '33101', '2020-06-30', 'Individual', 'Medium', 670, 65000.00, TRUE),

(1007, 'David', 'Garcia', 'david.garcia@email.com', '555-0107', '1975-04-25', '147 Cedar Ln', 'Houston', 'TX', 'USA', '77001', '2020-08-15', 'Individual', 'High', 780, 180000.00, TRUE),

(1008, 'Lisa', 'Rodriguez', 'lisa.r@email.com', '555-0108', '1987-06-18', '258 Birch Dr', 'Phoenix', 'AZ', 'USA', '85001', '2020-09-22', 'Individual', 'Medium', 705, 92000.00, TRUE),

(1009, 'James', 'Wilson', 'james.wilson@email.com', '555-0109', '1980-10-05', '369 Maple Ct', 'Denver', 'CO', 'USA', '80201', '2020-10-10', 'Individual', 'Low', 690, 75000.00, TRUE),

(1010, 'Patricia', 'Lee', 'patricia.lee@email.com', '555-0110', '1995-02-28', '741 Spruce Way', 'Portland', 'OR', 'USA', '97201', '2020-11-18', 'Individual', 'Medium', 665, 62000.00, TRUE),

(1011, 'William', 'Taylor', 'william.t@email.com', '555-0111', '1983-08-12', '852 Walnut Pl', 'Austin', 'TX', 'USA', '78701', '2021-01-05', 'Individual', 'High', 740, 135000.00, TRUE),

(1012, 'Linda', 'Thomas', 'linda.thomas@email.com', '555-0112', '1991-03-20', '963 Ash Blvd', 'Atlanta', 'GA', 'USA', '30301', '2021-02-14', 'Individual', 'Medium', 685, 82000.00, TRUE),

(1013, 'Richard', 'Moore', 'richard.m@email.com', '555-0113', '1986-07-09', '159 Fir St', 'Dallas', 'TX', 'USA', '75201', '2021-03-22', 'Individual', 'Low', 700, 98000.00, TRUE),

(1014, 'Mary', 'Jackson', 'mary.jackson@email.com', '555-0114', '1979-11-30', '357 Cypress Ave', 'Philadelphia', 'PA', 'USA', '19101', '2021-04-18', 'Business', 'Medium', 730, 200000.00, TRUE),

(1015, 'Thomas', 'White', 'thomas.white@email.com', '555-0115', '1993-01-16', '456 Redwood Dr', 'San Diego', 'CA', 'USA', '92101', '2021-05-12', 'Individual', 'High', 760, 145000.00, TRUE);

-- Insert Branches

INSERT INTO Branches VALUES

(1, 'NYC001', 'New York Main Branch', '500 Fifth Ave', 'New York', 'NY', 'USA', '555-2001', 'Robert Smith', '2015-01-15', 'Main', 1500, TRUE),

(2, 'SF001', 'San Francisco Downtown', '100 Market St', 'San Francisco', 'CA', 'USA', '555-2002', 'Maria Chen', '2016-03-20', 'Main', 1200, TRUE),

(3, 'CHI001', 'Chicago Loop Branch', '200 State St', 'Chicago', 'IL', 'USA', '555-2003', 'James Brown', '2016-06-10', 'Main', 980, TRUE),

(4, 'BOS001', 'Boston Financial District', '75 State St', 'Boston', 'MA', 'USA', '555-2004', 'Sarah Johnson', '2017-02-15', 'Main', 850, TRUE),

(5, 'SEA001', 'Seattle Downtown', '400 Pike St', 'Seattle', 'WA', 'USA', '555-2005', 'David Lee', '2017-08-22', 'Regional', 720, TRUE);

```
-- Insert Employees
INSERT INTO Employees VALUES
(2001, 'Robert', 'Smith', 'robert.smith@financehub.com', '555-3001', '2015-01-15', 'Branch Manager', 'Operations', 1, NULL, 125000.00, 4.5, TRUE),
(2002, 'Maria', 'Chen', 'maria.chen@financehub.com', '555-3002', '2016-03-20', 'Branch Manager', 'Operations', 2, NULL, 120000.00, 4.7, TRUE),
(2003, 'James', 'Brown', 'james.brown@financehub.com', '555-3003', '2016-06-10', 'Branch Manager', 'Operations', 3, NULL, 118000.00, 4.3, TRUE),
(2004, 'Sarah', 'Johnson', 'sarah.johnson@financehub.com', '555-3004', '2017-02-15', 'Branch Manager', 'Operations', 4, NULL, 115000.00, 4.6, TRUE),
(2005, 'David', 'Lee', 'david.lee@financehub.com', '555-3005', '2017-08-22', 'Branch Manager', 'Operations', 5, NULL, 110000.00, 4.4, TRUE),
(2006, 'Jennifer', 'Wilson', 'jennifer.wilson@financehub.com', '555-3006', '2018-01-10', 'Loan Officer', 'Lending', 1, 2001, 75000.00, 4.2, TRUE),
(2007, 'Michael', 'Davis', 'michael.davis@financehub.com', '555-3007', '2018-03-15', 'Financial Advisor', 'Wealth Management', 1, 2001, 85000.00, 4.8, TRUE),
(2008, 'Amanda', 'Martinez', 'amanda.m@financehub.com', '555-3008', '2018-06-20', 'Customer Service Rep', 'Customer Service', 2, 2002, 45000.00, 4.1, TRUE),
(2009, 'Christopher', 'Garcia', 'chris.garcia@financehub.com', '555-3009', '2019-01-12', 'Fraud Analyst', 'Risk Management', 1, 2001, 68000.00, 4.5, TRUE),
(2010, 'Jessica', 'Rodriguez', 'jessica.r@financehub.com', '555-3010', '2019-04-18', 'Investment Specialist', 'Wealth Management', 2, 2002, 92000.00, 4.7, TRUE),
(2011, 'Daniel', 'Taylor', 'daniel.t@financehub.com', '555-3011', '2019-08-22', 'Credit Analyst', 'Lending', 3, 2003, 72000.00, 4.3, TRUE),
(2012, 'Emily', 'Thomas', 'emily.thomas@financehub.com', '555-3012', '2020-01-15', 'Customer Service Rep', 'Customer Service', 1, 2001, 47000.00, 4.6, TRUE),
(2013, 'Matthew', 'Moore', 'matthew.moore@financehub.com', '555-3013', '2020-05-20', 'Loan Officer', 'Lending', 2, 2002, 78000.00, 4.4, TRUE),
(2014, 'Ashley', 'Jackson', 'ashley.j@financehub.com', '555-3014', '2020-09-10', 'Financial Advisor', 'Wealth Management', 4, 2004, 88000.00, 4.5, TRUE),
(2015, 'Joshua', 'White', 'joshua.white@financehub.com', '555-3015', '2021-02-14', 'Fraud Analyst', 'Risk Management', 2, 2002, 70000.00, 4.2, TRUE);

-- Insert Accounts
INSERT INTO Accounts VALUES
(5001, 1001, 'ACC10010001', 'Savings', 'Active', '2020-01-15', 45000.00, 'USD', 2.50, 1000.00, 'NYC001', '2024-03-15'),
(5002, 1001, 'ACC10010002', 'Checking', 'Active', '2020-01-15', 12500.00, 'USD', 0.00, 0.00, 'NYC001', '2024-03-14'),
(5003, 1002, 'ACC10020001', 'Savings', 'Active', '2020-02-20', 28000.00, 'USD', 2.50, 1000.00, 'SF001', '2024-03-10'),
(5004, 1003, 'ACC10030001', 'Checking', 'Active', '2020-03-18', 18500.00, 'USD', 0.00, 0.00, 'CHI001', '2024-03-12'),
(5005, 1003, 'ACC10030002', 'Money Market', 'Active', '2020-06-22', 75000.00, 'USD', 3.25, 5000.00, 'CHI001', '2024-03-08'),
(5006, 1004, 'ACC10040001', 'Savings', 'Active', '2020-04-12', 95000.00, 'USD', 2.75, 1000.00, 'BOS001', '2024-03-16'),
(5007, 1004, 'ACC10040002', 'Checking', 'Active', '2020-04-12', 22000.00, 'USD', 0.00, 0.00, 'BOS001', '2024-03-14'),
(5008, 1005, 'ACC10050001', 'Savings', 'Active', '2020-05-25', 35000.00, 'USD', 2.50, 1000.00, 'SEA001', '2024-03-11'),
(5009, 1006, 'ACC10060001', 'Checking', 'Active', '2020-06-30', 8500.00, 'USD', 0.00, 0.00, 'NYC001', '2024-03-13'),
```

(5010, 1007, 'ACC10070001', 'Money Market', 'Active', '2020-08-15', 125000.00, 'USD', 3.50, 5000.00, 'NYC001', '2024-03-15'),
(5011, 1008, 'ACC10080001', 'Savings', 'Active', '2020-09-22', 42000.00, 'USD', 2.50, 1000.00, 'SF001', '2024-03-09'),
(5012, 1009, 'ACC10090001', 'Checking', 'Active', '2020-10-10', 15000.00, 'USD', 0.00, 0.00, 'CHI001', '2024-03-12'),
(5013, 1010, 'ACC10100001', 'Savings', 'Active', '2020-11-18', 18000.00, 'USD', 2.50, 1000.00, 'SEA001', '2024-03-10'),
(5014, 1011, 'ACC10110001', 'Savings', 'Active', '2021-01-05', 68000.00, 'USD', 2.75, 1000.00, 'BOS001', '2024-03-14'),
(5015, 1012, 'ACC10120001', 'Checking', 'Active', '2021-02-14', 11000.00, 'USD', 0.00, 0.00, 'SF001', '2024-03-11');

-- Insert Transactions (Sample 30 transactions)
INSERT INTO Transactions VALUES
(10001, 5001, 'Deposit', '2024-01-05 10:15:00', 5000.00, 50000.00, 'Salary Credit', 'TXN10001', 'Completed', 'Online Banking', 'Income'),
(10002, 5002, 'Withdrawal', '2024-01-06 14:30:00', 500.00, 12000.00, 'ATM Withdrawal', 'TXN10002', 'Completed', 'ATM', 'Cash'),
(10003, 5001, 'Transfer Out', '2024-01-10 09:45:00', 2000.00, 48000.00, 'Bill Payment', 'TXN10003', 'Completed', 'Online Banking', 'Bills'),
(10004, 5003, 'Deposit', '2024-01-12 11:20:00', 3500.00, 31500.00, 'Freelance Payment', 'TXN10004', 'Completed', 'Wire Transfer', 'Income'),
(10005, 5004, 'Withdrawal', '2024-01-15 16:10:00', 1200.00, 17300.00, 'Grocery Shopping', 'TXN10005', 'Completed', 'Debit Card', 'Shopping'),
(10006, 5006, 'Deposit', '2024-01-18 10:30:00', 8000.00, 103000.00, 'Bonus Credit', 'TXN10006', 'Completed', 'Direct Deposit', 'Income'),
(10007, 5002, 'Transfer In', '2024-01-20 13:45:00', 1500.00, 13500.00, 'From Savings', 'TXN10007', 'Completed', 'Online Banking', 'Transfer'),
(10008, 5008, 'Withdrawal', '2024-01-22 15:20:00', 800.00, 34200.00, 'Restaurant', 'TXN10008', 'Completed', 'Debit Card', 'Dining'),
(10009, 5010, 'Deposit', '2024-01-25 09:15:00', 10000.00, 135000.00, 'Investment Returns', 'TXN10009', 'Completed', 'Wire Transfer', 'Investment'),
(10010, 5011, 'Withdrawal', '2024-01-28 12:40:00', 2500.00, 39500.00, 'Rent Payment', 'TXN10010', 'Completed', 'Check', 'Housing'),
(10011, 5001, 'Withdrawal', '2024-02-02 11:30:00', 300.00, 47700.00, 'Utility Bills', 'TXN10011', 'Completed', 'Online Banking', 'Bills'),
(10012, 5005, 'Deposit', '2024-02-05 14:15:00', 5000.00, 80000.00, 'Business Revenue', 'TXN10012', 'Completed', 'Wire Transfer', 'Income'),
(10013, 5007, 'Withdrawal', '2024-02-08 10:20:00', 1500.00, 20500.00, 'Medical Expenses', 'TXN10013', 'Completed', 'Debit Card', 'Healthcare'),
(10014, 5009, 'Deposit', '2024-02-10 16:45:00', 2000.00, 10500.00, 'Refund', 'TXN10014', 'Completed', 'ACH', 'Refund'),
(10015, 5012, 'Withdrawal', '2024-02-12 13:30:00', 600.00, 14400.00, 'Subscription Services', 'TXN10015', 'Completed', 'Auto Debit', 'Entertainment'),
(10016, 5014, 'Deposit', '2024-02-15 09:50:00', 7000.00, 75000.00, 'Salary Credit', 'TXN10016', 'Completed', 'Direct Deposit', 'Income'),
(10017, 5003, 'Withdrawal', '2024-02-18 15:10:00', 900.00, 30600.00, 'Fuel', 'TXN10017', 'Completed', 'Debit Card', 'Transportation'),
(10018, 5006, 'Transfer Out', '2024-02-20 11:25:00', 15000.00, 88000.00, 'Investment Transfer', 'TXN10018', 'Completed', 'Online Banking', 'Investment'),
(10019, 5001, 'Deposit', '2024-02-22 10:15:00', 4000.00, 51700.00, 'Dividend Income', 'TXN10019', 'Completed', 'ACH', 'Income'),

(10020, 5008, 'Withdrawal', '2024-02-25 14:40:00', 1200.00, 33000.00, 'Insurance Premium', 'TXN10020', 'Completed', 'Auto Debit', 'Insurance'),
(10021, 5013, 'Deposit', '2024-02-28 09:30:00', 2500.00, 20500.00, 'Bonus', 'TXN10021', 'Completed', 'Direct Deposit', 'Income'),
(10022, 5004, 'Withdrawal', '2024-03-01 16:20:00', 450.00, 16850.00, 'Clothing Purchase', 'TXN10022', 'Completed', 'Debit Card', 'Shopping'),
(10023, 5010, 'Withdrawal', '2024-03-03 11:15:00', 3000.00, 132000.00, 'Home Maintenance', 'TXN10023', 'Completed', 'Check', 'Housing'),
(10024, 5002, 'Deposit', '2024-03-05 13:45:00', 1000.00, 14500.00, 'Cashback Credit', 'TXN10024', 'Completed', 'ACH', 'Income'),
(10025, 5011, 'Withdrawal', '2024-03-08 10:30:00', 2000.00, 37500.00, 'Tuition Fee', 'TXN10025', 'Completed', 'Wire Transfer', 'Education'),
(10026, 5015, 'Deposit', '2024-03-10 15:20:00', 3000.00, 14000.00, 'Project Payment', 'TXN10026', 'Completed', 'Wire Transfer', 'Income'),
(10027, 5005, 'Deposit', '2024-03-12 09:40:00', 6000.00, 86000.00, 'Interest Credit', 'TXN10027', 'Completed', 'Auto Credit', 'Income'),
(10028, 5007, 'Withdrawal', '2024-03-14 14:10:00', 800.00, 21200.00, 'Pet Care', 'TXN10028', 'Completed', 'Debit Card', 'Personal'),
(10029, 5009, 'Withdrawal', '2024-03-15 11:50:00', 350.00, 10150.00, 'Phone Bill', 'TXN10029', 'Completed', 'Auto Debit', 'Bills'),
(10030, 5014, 'Withdrawal', '2024-03-16 16:35:00', 1500.00, 73500.00, 'Travel Booking', 'TXN10030', 'Completed', 'Online Banking', 'Travel');

-- Insert Loans
INSERT INTO Loans VALUES
(3001, 1001, 'Home Loan', 350000.00, 6.75, 240, 2625.00, 325000.00, '2021-06-15', '2041-06-15', 'Active', 'Property', 450000.00, 'Home Purchase'),
(3002, 1003, 'Personal Loan', 50000.00, 10.50, 60, 1068.00, 42000.00, '2022-03-10', '2027-03-10', 'Active', 'None', 0.00, 'Debt Consolidation'),
(3003, 1004, 'Auto Loan', 45000.00, 7.25, 60, 898.00, 32000.00, '2022-08-20', '2027-08-20', 'Active', 'Vehicle', 52000.00, 'Car Purchase'),
(3004, 1007, 'Business Loan', 200000.00, 8.50, 84, 3250.00, 175000.00, '2021-10-05', '2028-10-05', 'Active', 'Commercial Property', 280000.00, 'Business Expansion'),
(3005, 1008, 'Education Loan', 80000.00, 6.00, 120, 888.00, 65000.00, '2022-01-15', '2032-01-15', 'Active', 'None', 0.00, 'Higher Education'),
(3006, 1011, 'Home Loan', 425000.00, 6.50, 300, 2688.00, 410000.00, '2021-08-22', '2046-08-22', 'Active', 'Property', 550000.00, 'Home Purchase'),
(3007, 1014, 'Business Loan', 150000.00, 9.00, 72, 2450.00, 125000.00, '2022-04-18', '2028-04-18', 'Active', 'Equipment', 180000.00, 'Equipment Purchase'),
(3008, 1002, 'Personal Loan', 30000.00, 11.00, 48, 778.00, 18000.00, '2023-02-10', '2027-02-10', 'Active', 'None', 0.00, 'Home Renovation'),
(3009, 1006, 'Auto Loan', 35000.00, 7.75, 60, 706.00, 28000.00, '2022-09-15', '2027-09-15', 'Active', 'Vehicle', 40000.00, 'Car Purchase'),
(3010, 1009, 'Personal Loan', 25000.00, 10.00, 36, 806.00, 15000.00, '2023-06-01', '2026-06-01', 'Active', 'None', 0.00, 'Medical Emergency');

-- Insert LoanPayments (Sample payments)
INSERT INTO LoanPayments VALUES
(4001, 3001, '2024-01-15', 2625.00, 875.00, 1750.00, 324125.00, 'Paid', 'Auto Debit', 0.00),
(4002, 3001, '2024-02-15', 2625.00, 880.00, 1745.00, 323245.00, 'Paid', 'Auto Debit', 0.00),
(4003, 3001, '2024-03-15', 2625.00, 885.00, 1740.00, 322360.00, 'Paid', 'Auto Debit', 0.00),
(4004, 3002, '2024-01-10', 1068.00, 630.00, 438.00, 41370.00, 'Paid', 'Online Transfer', 0.00),
(4005, 3002, '2024-02-10', 1068.00, 635.00, 433.00, 40735.00, 'Paid', 'Online Transfer', 0.00),
(4006, 3002, '2024-03-10', 1068.00, 640.00, 428.00, 40095.00, 'Paid', 'Online Transfer', 0.00),

(4007, 3003, '2024-01-20', 898.00, 705.00, 193.00, 31295.00, 'Paid', 'Auto Debit', 0.00),
(4008, 3003, '2024-02-20', 898.00, 710.00, 188.00, 30585.00, 'Paid', 'Auto Debit', 0.00),
(4009, 3003, '2024-03-20', 898.00, 715.00, 183.00, 29870.00, 'Paid', 'Auto Debit', 0.00),
(4010, 3004, '2024-01-05', 3250.00, 2000.00, 1250.00, 173000.00, 'Paid', 'Wire Transfer', 0.00),
(4011, 3004, '2024-02-05', 3250.00, 2015.00, 1235.00, 170985.00, 'Paid', 'Wire Transfer', 0.00),
(4012, 3004, '2024-03-05', 3250.00, 2030.00, 1220.00, 168955.00, 'Paid', 'Wire Transfer', 0.00),
(4013, 3005, '2024-01-15', 888.00, 563.00, 325.00, 64437.00, 'Paid', 'Auto Debit', 0.00),
(4014, 3005, '2024-02-15', 888.00, 566.00, 322.00, 63871.00, 'Paid', 'Auto Debit', 0.00),
(4015, 3005, '2024-03-15', 888.00, 569.00, 319.00, 63302.00, 'Paid', 'Auto Debit', 0.00),
(4016, 3006, '2024-01-22', 2688.00, 1465.00, 1223.00, 408535.00, 'Paid', 'Auto Debit', 0.00),
(4017, 3006, '2024-02-22', 2688.00, 1473.00, 1215.00, 407062.00, 'Paid', 'Auto Debit', 0.00),
(4018, 3006, '2024-03-22', 2688.00, 1481.00, 1207.00, 405581.00, 'Paid', 'Auto Debit', 0.00),
(4019, 3008, '2024-02-10', 778.00, 513.00, 265.00, 17487.00, 'Paid', 'Online Transfer', 0.00),
(4020, 3008, '2024-03-10', 778.00, 518.00, 260.00, 16969.00, 'Paid', 'Online Transfer', 0.00);

-- Insert CreditCards
INSERT INTO CreditCards VALUES
(6001, 1001, '4532-XXXX-XXXX-1234', 'Platinum', 50000.00, 42000.00, 8000.00, 18.99, '2020-02-01', '2025-02-01', 'Active', 12500, 499.00),
(6002, 1002, '5425-XXXX-XXXX-5678', 'Gold', 30000.00, 24500.00, 5500.00, 21.49, '2020-03-15', '2025-03-15', 'Active', 6800, 199.00),
(6003, 1003, '3782-XXXX-XXXX-9012', 'Platinum', 75000.00, 68000.00, 7000.00, 17.99, '2020-04-10', '2025-04-10', 'Active', 18200, 599.00),
(6004, 1004, '6011-XXXX-XXXX-3456', 'Signature', 100000.00, 88000.00, 12000.00, 16.99, '2020-05-20', '2025-05-20', 'Active', 24500, 799.00),
(6005, 1005, '4532-XXXX-XXXX-7890', 'Silver', 20000.00, 16500.00, 3500.00, 22.99, '2020-06-12', '2025-06-12', 'Active', 4200, 99.00),
(6006, 1007, '5425-XXXX-XXXX-2345', 'Platinum', 80000.00, 70000.00, 10000.00, 17.49, '2020-09-05', '2025-09-05', 'Active', 22000, 599.00),
(6007, 1008, '4532-XXXX-XXXX-6789', 'Gold', 40000.00, 35000.00, 5000.00, 20.99, '2020-10-18', '2025-10-18', 'Active', 8500, 199.00),
(6008, 1011, '3782-XXXX-XXXX-0123', 'Platinum', 60000.00, 52000.00, 8000.00, 18.49, '2021-01-22', '2026-01-22', 'Active', 14800, 499.00),
(6009, 1012, '5425-XXXX-XXXX-4567', 'Silver', 25000.00, 21000.00, 4000.00, 22.49, '2021-03-10', '2026-03-10', 'Active', 5500, 99.00),
(6010, 1015, '6011-XXXX-XXXX-8901', 'Gold', 45000.00, 38000.00, 7000.00, 19.99, '2021-06-15', '2026-06-15', 'Active', 10200, 199.00);

-- Insert CardTransactions (Sample 25 transactions)
INSERT INTO CardTransactions VALUES
(7001, 6001, '2024-01-05 14:25:00', 'Amazon.com', 'Online Shopping', 450.00, 'Purchase', 'Posted', FALSE, 45),
(7002, 6002, '2024-01-08 10:15:00', 'Walmart', 'Groceries', 180.00, 'Purchase', 'Posted', FALSE, 18),
(7003, 6001, '2024-01-12 16:30:00', 'Delta Airlines', 'Travel', 850.00, 'Purchase', 'Posted', FALSE, 170),
(7004, 6003, '2024-01-15 11:45:00', 'Apple Store', 'Electronics', 1200.00, 'Purchase', 'Posted', FALSE, 240),
(7005, 6004, '2024-01-18 09:20:00', 'Hilton Hotels', 'Travel', 650.00, 'Purchase', 'Posted', FALSE, 130),
(7006, 6005, '2024-01-20 14:10:00', 'Shell Gas Station', 'Fuel', 75.00, 'Purchase', 'Posted', FALSE, 8),
(7007, 6006, '2024-01-22 15:40:00', 'Best Buy', 'Electronics', 920.00, 'Purchase', 'Posted', FALSE, 184),
(7008, 6007, '2024-01-25 12:30:00', 'Target', 'Shopping', 235.00, 'Purchase', 'Posted', FALSE, 24),
(7009, 6001, '2024-01-28 10:50:00', 'Starbucks', 'Dining', 35.00, 'Purchase', 'Posted', FALSE, 7),
(7010, 6008, '2024-02-02 16:15:00', 'Costco', 'Groceries', 380.00, 'Purchase', 'Posted', FALSE, 38),
(7011, 6003, '2024-02-05 11:25:00', 'Home Depot', 'Home Improvement', 560.00, 'Purchase', 'Posted', FALSE, 112),

(7012, 6002, '2024-02-08 14:40:00', 'Uber', 'Transportation', 45.00, 'Purchase', 'Posted', FALSE, 5),
(7013, 6004, '2024-02-10 09:30:00', 'Marriott Hotels', 'Travel', 720.00, 'Purchase', 'Posted', FALSE, 144),
(7014, 6009, '2024-02-12 15:20:00', 'CVS Pharmacy', 'Healthcare', 125.00, 'Purchase', 'Posted', FALSE, 13),
(7015, 6010, '2024-02-15 12:10:00', 'Nike Store', 'Apparel', 290.00, 'Purchase', 'Posted', FALSE, 58),
(7016, 6001, '2024-02-18 10:45:00', 'Whole Foods', 'Groceries', 165.00, 'Purchase', 'Posted', FALSE, 17),
(7017, 6006, '2024-02-20 16:30:00', 'British Airways', 'Travel', 1450.00, 'Purchase', 'Posted', TRUE, 290),
(7018, 6007, '2024-02-22 11:15:00', 'Netflix', 'Entertainment', 15.99, 'Purchase', 'Posted', FALSE, 2),
(7019, 6003, '2024-02-25 14:50:00', 'Sephora', 'Beauty', 185.00, 'Purchase', 'Posted', FALSE, 37),
(7020, 6008, '2024-02-28 09:40:00', 'Chipotle', 'Dining', 42.00, 'Purchase', 'Posted', FALSE, 8),
(7021, 6002, '2024-03-02 15:25:00', 'Exxon', 'Fuel', 68.00, 'Purchase', 'Posted', FALSE, 7),
(7022, 6004, '2024-03-05 12:35:00', 'Nordstrom', 'Apparel', 485.00, 'Purchase', 'Posted', FALSE, 97),
(7023, 6001, '2024-03-08 10:20:00', 'Amazon.com', 'Online Shopping', 325.00, 'Purchase', 'Posted', FALSE, 65),
(7024, 6010, '2024-03-12 16:10:00', 'United Airlines', 'Travel', 680.00, 'Purchase', 'Posted', FALSE, 136),
(7025, 6005, '2024-03-15 11:30:00', 'Walgreens', 'Healthcare', 95.00, 'Purchase', 'Posted', FALSE, 10);

-- Insert Investments
INSERT INTO Investments VALUES
(8001, 1001, 'Stocks', 'Apple Inc (AAPL)', 100.000, 15000.00, 17500.00, '2021-03-15', NULL, 'Active', 'Medium', 16.67),
(8002, 1003, 'Mutual Fund', 'Vanguard S&P 500', 500.000, 25000.00, 32000.00, '2020-06-20', NULL, 'Active', 'Low', 28.00),
(8003, 1004, 'Bonds', 'US Treasury 10Y', 50000.000, 50000.00, 52500.00, '2021-01-10', '2031-01-10', 'Active', 'Low', 5.00),
(8004, 1007, 'Stocks', 'Microsoft (MSFT)', 150.000, 45000.00, 58500.00, '2021-08-05', NULL, 'Active', 'Medium', 30.00),
(8005, 1004, 'Stocks', 'Tesla (TSLA)', 75.000, 18000.00, 13500.00, '2022-02-15', NULL, 'Active', 'High', -25.00),
(8006, 1011, 'Mutual Fund', 'Fidelity Growth Fund', 800.000, 40000.00, 48000.00, '2021-05-22', NULL, 'Active', 'Medium', 20.00),
(8007, 1003, 'Bonds', 'Corporate Bond AAA', 30000.000, 30000.00, 32100.00, '2022-03-10', '2027-03-10', 'Active', 'Low', 7.00),
(8008, 1015, 'Stocks', 'Amazon (AMZN)', 80.000, 24000.00, 27200.00, '2021-11-18', NULL, 'Active', 'Medium', 13.33),
(8009, 1007, 'Real Estate Fund', 'Vanguard REIT Index', 600.000, 60000.00, 69000.00, '2020-09-25', NULL, 'Active', 'Medium', 15.00),
(8010, 1014, 'Stocks', 'Google (GOOGL)', 120.000, 36000.00, 43200.00, '2021-07-12', NULL, 'Active', 'Medium', 20.00),
(8011, 1001, 'Mutual Fund', 'BlackRock Equity Fund', 400.000, 20000.00, 23600.00, '2022-01-20', NULL, 'Active', 'Low', 18.00),
(8012, 1008, 'Bonds', 'Municipal Bond', 25000.000, 25000.00, 26250.00, '2022-06-15', '2032-06-15', 'Active', 'Low', 5.00),
(8013, 1011, 'Stocks', 'NVIDIA (NVDA)', 50.000, 22000.00, 35000.00, '2021-10-08', NULL, 'Active', 'High', 59.09),
(8014, 1015, 'Mutual Fund', 'T. Rowe Price Growth', 350.000, 17500.00, 21000.00, '2022-04-12', NULL, 'Active', 'Medium', 20.00),
(8015, 1012, 'Stocks', 'Meta (META)', 90.000, 27000.00, 32400.00, '2021-12-05', NULL, 'Active', 'High', 20.00);

-- Insert CustomerService tickets
INSERT INTO CustomerService VALUES
(9001, 1001, 2012, '2024-01-10 09:30:00', 'Account Access', 'Unable to login to online banking', 'High', 'Resolved', '2024-01-10 11:15:00', 5),
(9002, 1002, 2008, '2024-01-15 14:20:00', 'Card Issue', 'Credit card declined at merchant', 'Medium', 'Resolved', '2024-01-15 16:45:00', 4),

(9003, 1003, 2012, '2024-01-22 10:45:00', 'Transaction Dispute', 'Unauthorized transaction on account', 'High', 'Resolved', '2024-01-25 14:30:00', 3),
(9004, 1005, 2008, '2024-02-05 11:15:00', 'Loan Inquiry', 'Question about loan payment schedule', 'Low', 'Resolved', '2024-02-05 12:00:00', 5),
(9005, 1006, 2012, '2024-02-12 15:30:00', 'Account Balance', 'Discrepancy in account balance', 'Medium', 'Resolved', '2024-02-13 10:20:00', 4),
(9006, 1008, 2008, '2024-02-18 09:50:00', 'Card Request', 'Request for credit limit increase', 'Low', 'Resolved', '2024-02-20 14:15:00', 5),
(9007, 1009, 2012, '2024-02-25 13:40:00', 'Online Banking', 'Password reset request', 'Medium', 'Resolved', '2024-02-25 14:10:00', 5),
(9008, 1010, 2008, '2024-03-02 10:20:00', 'Transaction Dispute', 'Wrong amount debited', 'High', 'In Progress', NULL, NULL),
(9009, 1012, 2012, '2024-03-08 16:15:00', 'Card Issue', 'Lost credit card', 'High', 'Resolved', '2024-03-08 17:30:00', 5),
(9010, 1004, 2008, '2024-03-14 11:30:00', 'Investment Query', 'Portfolio performance inquiry', 'Low', 'Resolved', '2024-03-14 13:45:00', 4);

-- Insert FraudAlerts
INSERT INTO FraudAlerts VALUES
(9501, 1002, 5003, '2024-01-20 03:15:00', 'Unusual Transaction', 75, 'Large withdrawal at unusual hour', 'Resolved', 2009, '2024-01-20 09:30:00', FALSE),
(9502, 1006, 5009, '2024-02-05 23:45:00', 'Multiple Login Attempts', 85, 'Multiple failed login attempts from foreign IP', 'Resolved', 2009, '2024-02-06 10:15:00', TRUE),
(9503, 1003, NULL, '2024-02-15 02:30:00', 'Card Fraud', 92, 'Multiple card transactions in different countries', 'Resolved', 2015, '2024-02-15 11:20:00', TRUE),
(9504, 1009, 5012, '2024-02-28 04:20:00', 'Unusual Transaction', 68, 'Transaction amount significantly above average', 'Resolved', 2009, '2024-02-28 10:45:00', FALSE),
(9505, 1010, 5013, '2024-03-10 01:15:00', 'Account Takeover', 95, 'Suspicious changes to account information', 'Under Investigation', 2015, NULL, NULL);

-- Insert Tools
INSERT INTO Tools VALUES
(1, 'Tableau', 'Data Visualization', 840.00, 'Annual', '2023.3', TRUE),
(2, 'Python', 'Programming', 0.00, 'Open Source', '3.11', TRUE),
(3, 'SQL Server', 'Database', 1200.00, 'Annual', '2022', TRUE),
(4, 'Power BI', 'Data Visualization', 120.00, 'Monthly', 'Pro', TRUE),
(5, 'Excel', 'Spreadsheet', 150.00, 'Annual', '2021', TRUE);

-- Insert AnalystTools
INSERT INTO AnalystTools VALUES
(2007, 1, 'Expert', TRUE, '2024-03-15'),
(2007, 2, 'Advanced', TRUE, '2024-03-10'),
(2007, 3, 'Expert', TRUE, '2024-03-14'),
(2010, 1, 'Expert', TRUE, '2024-03-12'),
(2010, 4, 'Advanced', FALSE, '2024-03-08'),
(2011, 3, 'Expert', TRUE, '2024-03-16'),
(2011, 5, 'Advanced', FALSE, '2024-03-11');

-- Insert TrainingPrograms
INSERT INTO TrainingPrograms VALUES
(501, 'Advanced SQL Analytics', 'Technical', 40, 799.00, 'DataCamp', '2024-01-10', '2024-02-10'),
(502, 'Financial Risk Management', 'Finance', 32, 1200.00, 'CFA Institute', '2024-02-01', '2024-03-01'),
(503, 'Python for Finance', 'Technical', 48, 899.00, 'Coursera', '2024-01-15', '2024-03-15'),

(504, 'Customer Service Excellence', 'Soft Skills', 16, 299.00, 'Internal', '2024-02-15', '2024-02-20'),
(505, 'Fraud Detection Techniques', 'Risk Management', 24, 650.00, 'ACFE', '2024-03-01', '2024-03-20');

-- Insert AnalystTraining
INSERT INTO AnalystTraining VALUES
(10001, 2007, 501, '2024-02-10', 95, TRUE),
(10002, 2010, 502, '2024-03-01', 88, TRUE),
(10003, 2011, 501, '2024-02-10', 92, TRUE),
(10004, 2008, 504, '2024-02-20', 90, TRUE),
(10005, 2009, 505, '2024-03-20', 87, TRUE),
(10006, 2012, 504, '2024-02-20', 85, TRUE),
(10007, 2014, 503, '2024-03-15', 91, TRUE),
(10008, 2015, 505, '2024-03-20', 89, TRUE);

Part 2: Sql Challenges (25 Questions)

Instructions: Write SQL queries to answer the following questions.

Question 1:
List all customers who have a credit score above 700 and an annual income greater than $80,000. Display customer name (first and last combined), credit score, annual income, and risk profile. Sort by credit score in descending order.

Question 2:
Calculate the total transaction amount for each transaction category in 2024. Only include categories where the total transaction amount exceeds $5,000. Show category name and total amount, sorted by amount in descending order.

Question 3:
Display all active loans with customer details and current payment status. Show customer name, loan type, loan amount, outstanding balance, monthly EMI, and the date of their last payment (if any). Sort by outstanding balance descending.

Question 4:
Find all customers whose total account balance (the sum of all their accounts) is greater than the average total balance across all customers.Display the customer name, total balance, and the number of accounts they have.

Question 5:
Create a customer segmentation report that categorises customers based on their total investment value: 'High Value' (>$100,000), 'Medium Value' ($50,000-$100,000), 'Low Value' ($20,000-$50,000), 'New Investor' (<$20,000). Show customer name, total investment value, and segment. Include only customers with investments.

Question 6:
Identify all customers who have accounts but have never taken any loans. Display customer ID, name, total account balance, and number of accounts.

Question 7:
List all employees along with their manager's name and department. Show employee name, job title, manager name, and department. Include employees without managers (CEO level).

Question 8:
Rank credit cards by their utilisation percentage (current_balance/credit_limit * 100) within each card type. Show card type, customer name, credit limit, current balance, utilisation percentage, and rank within card type.

Question 9:
Calculate monthly transaction volume and average transaction amount for each month in 2024. Show month, number of transactions, total amount, and average amount. Order by month chronologically.

Question 10:
Find the top 5 customers who have spent the most on credit card transactions in the 'Travel' category. Show customer name, total travel spending, number of travel transactions, and total reward points earned from travel.

Question 11:
List all accounts that have received deposits greater than $5,000 in the last 60 days. Show account number, account type, customer name, and current balance. Use EXISTS or IN with a subquery.

Question 12:
Create a unified report of all financial products a customer holds. For customer_id 1004, show: product type ('Account', 'Loan', 'Credit Card', 'Investment'), product identifier (account number, loan ID, card number, or investment type), and current value/balance. Use UNION to combine results.

Question 13:
For account 5001, show all transactions in 2024 with a running balance. Display transaction date, transaction type, amount, and running balance (calculated using a window function). Order by transaction date.

Question 14:
Analyse loan performance by loan type. Show loan type, number of loans, total disbursed amount, total outstanding balance, average interest rate, and payment completion percentage (total paid / total disbursed).

Question 15:
For each branch, show the branch name, the total number of employees, total number of customers and the percentage of customers relative to total bank customers.

Question 16:
Find customers who have made more than 5 transactions and whose average transaction amount is greater than $500. Show customer name, number of transactions, total transaction amount, and average transaction amount.

Question 17:
Track the month-over-month change in total loan disbursements. Show the month, total amount disbursed, previous month's amount, and percentage change. Calculate for all months where loans were disbursed.

Question 18:
Create a fraud risk assessment report showing customers with high-risk indicators: More than 3 fraud alerts OR any confirmed fraud cases OR credit card utilisation> 80%. Show customer name, number of fraud alerts, confirmed fraud count, max credit card utilisation %, and overall risk level (High/Medium/Low).

Question 19:
Find customers who have both a credit card AND an active loan, but do NOT have any investments. Use appropriate set operations or equivalent logic. Display customer ID, name, email, and their risk profile.

Question 20:
Write a query to find all transactions for a specific account within a date range that would benefit from indexing. Then, suggest the CREATE INDEX statement that would optimize this query. Show the query and index suggestion as comments. **Query:** Find all transactions for account 5001 between '2024-01-01' and '2024-03-31' where amount > 1000, sorted by transaction_date.

Question 21
Using CTEs, create a hierarchical report showing the employee management structure with levels (CEO = Level 1, direct reports = Level 2, etc.). Show employee name, manager name, level, and department.

Question 22
Calculate the percentile rank of each customer's total account balance within their risk profile group. Show customer name, risk profile, total balance, and percentile rank.

Question 23
Write a query that could be converted to a stored procedure to calculate the total monthly EMI obligations for a given customer_id. Include all active loans and show the loan type, EMI amount, and total. Comment on how you would convert this to a stored procedure.

Question 24
Write a query to identify data integrity issues:
-Accounts with negative balances
-Loans with outstanding balance > original loan amount
-Credit cards with available credit < 0
-Transactions where balance_after doesn't match the expected calculation

Show issue type, affected record ID, and details.

Question 25
Create a comprehensive dashboard query that shows:
-Total customers by risk profile
-Total deposits, withdrawals, and net flow for the current month
-Average credit card utilisation by card type
-Loan portfolio health (% of on-time payments)
-Top 3 transaction categories by volume
-Use multiple subqueries or CTEs to build this report.

Submission Guidelines

To successfully complete this coursework
1. WRITE SQL QUERIES:
- Write correct SQL queries for all 25 questions
- Ensure queries are properly formatted and commented
- Test each query and verify the results
- Use proper naming conventions and indentation

2. EXPORT YOUR WORK:
- Save your SQL script file (.sql) as: YourName_FinanceHub_Assignment.sql and upload on educollab as a single .sql file.